

## DPS912 Lab 1: Monitoring Process Memory Usage

**Due: Thursday, May 14, 2026 (11:59pm)**

Information on all processes running on a Linux system is contained within the **/proc** directory. All directories under **/proc** which are digits represent a process. If you issue the command **ps aux** you will see all processes that are running. You can also see the processes ordered by cpu usage by issuing the command **top**.

In this lab, you will write a C/CPP program to scan all processes running on the system and report on those which have used more than 10Mbytes of resident memory. You can do this by going through each process directory in **/proc** and looking at the file **status**. For instance, let's say you want to look at the status information for process 10167:

```
$ cd /proc/10167
```

```
$ vim status
```

Inside the file **status** are variables such as **Name:** (the process name) , **Pid:** (the process pid), **VmSize:** size of the process's data segment (its heap + other anonymous writable mappings) and Resident memory is represented by **VmRSS:**

1. Write and execute a C/CPP program that will scan every directory in **/proc** that represents a process (each directory that begins with a number represents a process). Then prepare the following outputs:
  - **Output1:** List all Pid, State, VmSize, VmRSS of all accessible process in **/proc/[pid]/status**.
  - **Output2:** Filter processes with high value of VmRSS and Print the **top 5** ( print out the process name, the process id and the memory used). See the code at the following link as an example on how to get all files in a directory:  
<https://www.unix.com/programming/33263-unix-c-get-files-folder.html>.
  - Note: Walk **/proc**, identify numeric PIDs, and **gracefully** skip those you cannot read (handle EACCES, races).
2. Open a second command line and execute the command **top** which orders processes by cpu usage.

```
$ top
```

Compare the processes reported in **top** with those reported by your program. You can cancel the execution of **top** with **ctrl-c**.  
See the following link for an explanation of top:  
<https://linux.die.net/man/1/top>.
3. Explain the difference between virtual memory and resident memory. What is virtual memory? What is resident memory?

4. Run the program on Matrix and also in your local Linux environment. Research and report on any of the four of the processes returned by your program (two from Matrix, two from local Linux env). Simply explain what these processes do.

The following web-page contains detailed information about each of the files under **/proc**:  
 proc - process information pseudo-filesystem: <https://linux.die.net/sag/proc-fs.html>.

Sample source code that retrieves system information such as model name, cpu speed, memory usage, processes running/blocked, up time and idle time can be seen at [MemMonitor.cpp](#).

## Assignment Submission:

- Complete all steps, Add all output-screenshot and explanations (if required) to a MS-Word file.
- Add the following declaration at the top of MSWORD file
 

```

/*****
***
* DPS912-Lab1
* I declare that this lab is my own work in accordance with Seneca Academic Policy.
* No part of this assignment has been copied manually or electronically from any other source
* (including web sites) or distributed to other students.
*
* Name: _____ Student ID: _____ Date: _____
*
*
*****
**/
      
```
- Please submit the Source code (zip all .c, .h, and makeFiles)
- Please answer the following two declarations:
  - **D1)** On a scale from 1 to 5, **How much did you use generative AI to complete this assignment?**
    - where:
    - **1** means you did not use generative AI at all
    - **2** means you used it very minimally
    - **3** means you used it moderately
    - **4** means you used it significantly
    - **5** means you relied on it almost entirely
    - **Your answer :**
  - **D2)** On a scale from 1 to 5, **How confident are you in your understanding of the generative AI support you utilized in this assignment, and in your ability to explain it if questioned?**

- where:
  - **1** means "Not confident at all – I do not understand the generative AI support I used and cannot explain it."
  - **2** means "Slightly confident – I understand a little, but I have many uncertainties."
  - **3** means "Moderately confident – I understand the majority of the support, though some parts are unclear."
  - **4** means "Very confident – I understand most of the AI support well and can explain it with minor gaps."
  - **5** means "Extremely confident – I fully understand the generative AI support I used and can clearly explain or justify it if asked."
  - **Your answer :**
- Please submit the Source code (zip all .c, .h, and makeFiles)

## Important Note:

- **LATE SUBMISSIONS for labs.** There is a deduction of 10% for Late assignment submissions, and after three days it will grade of zero (0).
  - Please submit the incomplete work before deadline in order to be eligible for any form of extension/accommodation.
- This lab should be submitted along with a video-recording which contains a detailed walkthrough of solution. Without recording, the assignment can get a maximum of 1/3 of the total.
  - Note: In case you are running out of time to record the video, you can submit the assignment (source code + screenshots) by the deadline and submit the video within 24 hours after the deadline.